Electronics and signals for a robot sensor node

M4 • 27 Dec 2026 - 31 Jan 2027 • 75 hours

What you will build

M04 delivers a **low-voltage electrical experiment log and a sensor-signal filtering report**. The first part teaches assembly and measurement of simple future-robot circuits: a status indicator, analogue-input divider, digital contact and a switch for a small inductive load. The second explains why functioning electronics do not guarantee usable data: sampling rate, quantisation, aliasing and delay change the observed motion.

These tasks directly support leg and hand design. A foot-contact sensor or limit switch must produce a defined logic level. An analogue position sensor requires a suitable voltage range. A body-mounted IMU sees both useful tilt and vibration. Filtering may suppress jitter, but delayed angles can degrade later feedback. Measure this tradeoff in M04 before programming the M06 controller.

Equipment and monthly scope

You need an adjustable supply with current limiting and a suitable low-voltage output, a multimeter, breadboard, wires, resistors, LEDs, small capacitors, a switch and switching-circuit components. Frequency experiments may use simulation; an oscilloscope or logic analyser allows physical measurement but does not replace circuit understanding. First record available equipment specifications, then select compatible components.

Battery packs and high currents are outside the plan before checkpoint G1. Here, use bench circuits with limited stored energy; leg power drives, loaded motors and whole-robot power come later. A small relay coil with known specifications and unloaded contacts is sufficient for the inductive experiment. Derive current limits and fuse ratings from the actual components rather than copy universal values.

Preparation after M02-M03

Retain M03 noisy encoder and IMU recordings, timestamps and the 30-repeat protocol. Create circuits/, measurements/, signals/ and filter_report/. Link every measurement to a circuit revision, component values, probe locations and instrument mode. State which results were calculated, simulated or physically measured.

Initial check: distinguish voltage and current, convert mA to A and μF to F, and solve $I=V/R$ for a known resistor. If electronics is entirely new, devote the first two hours to circuit diagrams and the multimeter; use familiar Python for software work. Working ESP32 firmware is not yet required this month.

Allocating 75 hours and laboratory procedures

Four blocks

A, hours 1-20: 10 h theory + 10 h practice. Voltage, current, resistance, power and energy; Ohm's and Kirchhoff's laws; series and parallel branches; multimeter use. Deliverables: LED and loaded divider with predictions and measurements.

B, hours 21-40: 10+10 h. Capacitors, diodes, BJT/MOSFET and flyback diode; pull resistors, logic levels, ground and supply decoupling; wires, connectors, crimping and fuses. Deliverables: verified low-power switch and wiring diagram for a future sensor node.

C, hours 41-60: 10+10 h. Sampling, Nyquist, aliasing and quantisation; low-pass, high-pass and notch filters; moving average, IIR and complementary estimation. Deliverable: reproducible encoder and IMU processing with error checks.

D, hours 61-75: 5+10 h. Frequency response, bandwidth, phase, FFT, vibration spectrum, ADC/PWM and sensor delay. Deliverables: noise-versus-delay table, aliasing experiment and final presentation. Total: 35 h of analysis and 40 h of practice.

Scheduling sessions

Divide ten theory hours into five two-hour sessions and the final 5 h into 2+2+1. Divide each 10 h practical into five parts: circuit and prediction, assembly, measurement, edge cases and documentation. For electrical experiments, calculate and draw wiring first, assemble without power, then measure only after checking.

First-page dates follow the original plan. Boundary dates may overlap a neighbouring module; holidays, rest and integration stages are not recalculated. Work steps fill available sessions sequentially without double counting. Time spent merely waiting for recording or printing does not add study hours.

Required measurement sequence

Before power-up, check polarity, pinout, component limits, common return and accidental shorts. Measure voltage across a circuit section; measure current in series with its branch; measure resistance and continuity only on an unpowered circuit with discharged capacitors. After current measurement, return the probe to the voltage socket. Do not test incorrect ammeter connection by creating a real short circuit.

Before the physical inductive experiment, check the circuit against component documentation and, if inexperienced, review it with someone who can assemble such circuits. Without the necessary review or equipment, perform calculation and simulation and mark the physical work pending. A simulator screenshot cannot replace real-circuit acceptance. If behind schedule, reduce filter variants while retaining basic circuits and delay measurement.

Direct current: indicator and analogue input

Study before assembly

Voltage is potential difference; current is charge flow; resistance relates them in an ohmic element: $V=I \cdot R$. Power $P=V \cdot I$ is measured in watts; energy $E=P \cdot t$ at constant power is measured in joules. Kirchhoff's current law expresses current balance; the voltage law expresses loop-voltage balance. Choose arrow directions beforehand: a negative calculated value indicates the reverse direction.

Study series and parallel connections, open-circuit voltage and loading. An LED is not a constant resistor: its forward voltage depends on current and temperature. Calculate the limiting resistor for permissible current and the worst combination of supply voltage and forward drop, then check resistor power.

Task 1. Node status indicator

Input: supply and LED with manufacturer documentation, and two suitable resistors. A teaching calculation for 5 V, $V_f \approx 2$ V and 5 mA gives about 600 ohm; this illustrates arithmetic, while the actual value depends on your components and margin. Draw the loop, label voltage drops and current, then assemble LED and resistor with the supply disconnected.

After checking the circuit, enable power with the calculated limit. Measure supply, LED and resistor voltages; compute current as V_R/R and, if needed, compare with a correctly connected direct measurement. Deliverables: schematic, photograph, component values and a predicted-versus-measured table. Check voltage balance. Study reversed polarity and a missing resistor in a diagram or simulation without exposing components to damage.

Task 2. Divider as a position-sensor model

Build a divider from two known resistors. Predict $V_{out}=V_{in} \cdot R2/(R1+R2)$, then measure without an added load. Add a third resistor parallel to the lower leg and recalculate using equivalent resistance. This demonstrates the practical meaning of the next device's input resistance.

Compare three positions of a model resistive sensor; plot voltage against divider-leg ratio. Deliver an error table accounting for resistor and instrument tolerances. Edge case: divider output exceeds the future analogue-to-digital converter (ADC) input limit; identify incompatibility during calculation. Do not treat a divider as a complete voltage regulator or universal input protection.

Components, ground and logic levels

From passive components to switching

Study capacitors through $Q=C\cdot V$ and energy $C\cdot V^2/2$, and RC charging/discharging with time constant $\tau=R\cdot C$. A diode conducts mainly in one direction and has current and voltage limits. A BJT is controlled by base current; a MOSFET by gate-source voltage and gate charge. On-state losses and switching transients matter, beyond the component name.

Distinguish $V_{GS(th)}$ from the voltage at which the manufacturer specifies $R_{DS(on)}$: threshold does not mean a fully enhanced power switch. Study the selected package's exact pinout. For a digital input, compare source VOH/VOL with receiver VIH/VIL and absolute limits. A '3.3 V' board label does not make all pins 5 V compatible.

Task 3. Foot contact and pull-up resistor

Model the contact with a normal switch connecting the input to ground. A pull-up resistor connects the input to a suitable logic supply. Before attaching a controller, measure open/closed contact levels with a multimeter and calculate pull-up current. For teaching values 3.3 V and 10 kohm, closed-contact current is about 0.33 mA; verify the final value against the input and wire length.

Try two resistor values and compare prediction with measurement. A broken wire must have a defined logic meaning for future diagnostics, though this need not distinguish it from an open contact. Deliver a state table for 'open, closed, broken wire' and an explanation. First observe or simulate switch bounce; firmware debouncing comes in M06.

Task 4. RC circuits, decoupling and return current

Build an RC circuit in a simulator, apply a step and predict the time to approximately 63% of final voltage. With a suitable instrument, measure the transient in a circuit with little stored energy. Double R or C and compare the timescale. Save a plot and τ-estimation error; account for instrument input resistance.

Draw supply-current paths for the sensor and a separate load. Explain why decoupling capacitors sit near supply pins and why a long shared return wire can introduce interference. Do not disconnect ground while powered to 'demonstrate failure'. Model shared-wire resistance and show the added voltage drop caused by load current.

Protected switching and sensor-bench connections

Task 5. Small inductive load

Choose a low-power, low-voltage relay coil with known rated voltage, resistance and current. Leave relay contacts unloaded. Draw a low-side switch: coil between positive supply and drain or collector, source or emitter to common ground. Connect the flyback diode across the coil so it is reverse-biased in the normal energised state; on a standard positive rail, its cathode connects to the positive side.

First calculate coil current, switch losses, control-level compatibility and diode-rating margins. Study any ULN2003A option through its documentation: connect the clamp-diode common pin correctly, and do not assume the whole breadboard assembly can carry the IC's rated currents. With a MOSFET, ensure a defined off-state gate voltage and check RDS(on) at the actual drive voltage.

Experiment sequence and acceptance

1. Check the circuit on paper and in simulation. 2. Verify components and pinouts. 3. Assemble unpowered after selecting the supply current limit and fuse. 4. Start with static on/off operation. 5. Measure coil supply and switch voltage drop. 6. If an oscilloscope is available and its ground connection is understood, compare the measured transient with the model.

Deliver a log of currents, voltages, calculated losses and turn-off response. The coil must reliably de-energise, current must remain within component limits, and wiring must show no overload signs. Perform the no-flyback-diode experiment only in simulation. A multimeter cannot establish absence of a brief voltage spike; explicitly state if fast transients were not measured.

Task 6. Wiring, connector and fuse

List connection details: wire length and cross-section, contact type, connector current rating, retention method and labels. Calculate $\Delta V = I \cdot R$ across outgoing and return wires. Select fuse rating together with its time-current characteristic, operating and inrush currents, voltage and the weakest section's allowable conditions; 'slightly above current' is insufficient.

On an unpowered sample, inspect crimps visually and apply a light controlled pull according to contact instructions; check continuity of each conductor and isolation from adjacent ones. Prepare a pin→signal→voltage→wire_colour table and photographs of both ends. Edge cases: mirrored connector pinout and reversed polarity. Detect these before powering rather than rely on protection after an error.

Sampling: when a sensor reports different motion

Theory

Sampling converts a continuous signal into values at time intervals. fs=1/dt is measured in hertz. A band-limited signal requires a sampling rate above twice its highest frequency, plus practical margin for an antialiasing filter's transition band. Once frequencies have aliased, an ordinary digital low-pass filter applied after recording cannot reliably separate them.

ADC quantisation limits amplitude resolution; sampling limits temporal resolution. Study saturation, missing samples and uneven timestamp intervals separately. PWM is a pulse train with a defined on-time fraction; its frequency differs from command-update frequency and does not automatically produce a precise analogue voltage.

Task 7. Vibration becomes apparent motion

Generate ground truth on a dense time grid: a useful 2 Hz sinusoid plus 70 Hz vibration. Sample at 100 Hz and show the false 30 Hz component; repeat at 250 Hz. All numbers are teaching examples. Compare time plots and spectra, labelling true and observed frequencies.

Next, apply a suitable antialiasing filter to the dense recording before downsampling, and compare with 'downsample first, filter afterwards'. Deliver three spectra and explain operation order. Edge case: a sinusoid at fs/2 with an unsuitable phase; a few samples cannot reliably recover its amplitude. Do not test the Nyquist criterion only at the exact boundary.

Task 8. ADC and actual timestamps

Model conversion of voltage 0…Vref to an N-bit ADC integer code with explicit rounding and saturation rules. Calculate the quantisation step for the chosen model, plot error during a slow voltage change and check the smallest distinguishable change in a model joint position.

Add timestamp-interval variation (jitter) and one missing sample. Compare angle derivatives using constant dt and actual intervals; incorporate M01 checks. Deliver a CSV with raw_code, t_s, voltage_V, angle_rad and flags. Do not silently treat irregular samples as uniform before FFT: for this laboratory, select a uniform segment or explicitly resample onto a grid and describe the procedure.

Filters for encoders and body tilt

Required concepts

A low-pass filter preserves slow components and attenuates high frequencies; a high-pass filter attenuates DC and slow components; a notch filter suppresses a narrow band. A moving average uses a finite sample window; an IIR uses previous state. A simple low-pass filter is $y[k]=\alpha\cdot x[k]+(1-\alpha)\cdot y[k-1]$, with $0<\alpha\leq 1$. Smaller α gives stronger smoothing and slower response.

Study causality: a real-time filter cannot access future samples. Forward-backward processing such as filtfilt is useful for recorded-data analysis, but cannot establish a streaming algorithm's delay. Define initial filter state and behaviour after missing samples separately.

Task 9. Encoder noise versus delay

Input: a slow reference trajectory with a stationary section and a step, followed by a copy with noise and outliers. Implement causal moving averages with windows 3, 5, 11 and IIR filters with three α values. Calculate SD on the same stationary section, motion RMSE and time to reach 50% of the step relative to the reference. Distinguish step delay from 10-90% rise time.

Deliver a parameter table, time plots and a chosen tradeoff for the future joint monitor. Edge cases: first packet, NaN, a long gap and restart. NaN handling must allow valid IIR state to recover. State the policy for each case: reject, reset or continue with a flag; the policy must be visible in output data.

Task 10. Complementary tilt estimation

Generate a synthetic body-tilt trajectory in one plane with known $\theta(t)$. One input is angular velocity to integrate, with a small constant bias; the other is a noisy accelerometer angle estimate assuming small translational acceleration. Implement $\theta_{est}=\beta\cdot(\theta_{prev}+\omega\cdot dt)+(1-\beta)\cdot\theta_{acc}$ and check rad and s units. For the initial discrete approximation, use $\beta=\tau/(\tau+dt)$, with chosen time constant τ in seconds; recalculate β when dt changes and check the response.

Compare each separate estimate and their combination at rest, during tilt and during artificial linear acceleration. The last case must reveal the accelerometer-angle limitation. Do not add full 3D orientation, quaternions or automatic calibration here. Deliver drift/noise plots and describe conditions in which this simplified complementary filter is suitable.

Frequency response and latency budget

What a vibration spectrum means

FFT converts a finite, uniformly sampled recording into frequency components. Study recording length, frequency spacing fs/N, DC component, window functions and spectral leakage. Spectrum amplitude and power spectral density are different quantities; acceleration PSD has units $(m/s^2)^2/Hz$. Welch's method helps estimate a noise spectrum, but window parameters affect resolution.

A filter's frequency response describes amplitude and phase changes at each frequency. For a sinusoid, time delay relates to phase, but one delay constant may not describe an arbitrary signal. Do not approve a filter for future control from one smoothed plot: relate bandwidth and phase to task dynamics.

Task 11. Measure the frequency tradeoff

Pass a set of sinusoids through the task 9 causal filter. After the initial transient, measure amplitude ratio and phase shift; compare with scipy.signal.freqz. Add one synthetic vibration peak and calculate PSD with Welch's method. If using a notch filter, first specify centre frequency and suppressed bandwidth, then verify that nearby useful components remain intact.

Deliver amplitude/phase response plots and a three-variant table. For each setting, report stationary SD, motion RMSE, step delay and attenuation of the selected vibration. Define acceptance thresholds from the teaching monitor's requirements; the conclusion does not establish suitability for balancing a real robot.

From ADC to computed state

Draw the timing sequence: physical event → wait for sampling → conversion and internal sensor filter → transmission → processing → publication. List ADC sampling rate, output-message rate, PWM update rate and computation time separately. '100 Hz' does not describe the whole chain. Some times come from manufacturer documentation, some are measured and some remain unknown.

Add a known message delay to the test-bench model and compare physical-change time with filter-output time. If source and receiver clocks are not synchronised, their timestamp difference is not a reliable one-way latency. M04 needs one shared simulation timebase and a separate list of future M06-M07 hardware measurements.

Work sequence: hours 1-40

Block A: understand multimeter readings

Hours 1-2. Inventory equipment, study modes and convert units. **Hours 3-4.** Ohm's law, power and LED calculations. **Hours 5-6.** Kirchhoff's laws and series/parallel branches. **Hours 7-8.** Divider, loading and instrument input. **Hours 9-10.** Draw correct probe placement for all forthcoming measurements and check circuits.

Hours 11-20 - practice. 2 h: unpowered LED assembly and checks; 2 h: LED measurements and voltage balance; 2 h: unloaded and loaded divider; 2 h: repeat measurements and account for tolerances; 2 h: log, photographs and error review. Do not spend the practical buying equipment: prepare it before the relevant session.

At hour 20, choose a new resistor pair and predict divider output before measuring. Explain deviations from nominal-value calculations: tolerances, actual supply, instrument loading or connection errors. Identifying testable causes is more useful than adjusting an answer to an ideal number.

Block B: from contact to small coil

Hours 21-22. Capacitors, diodes and RC transients. **Hours 23-24.** BJT/MOSFET, pinouts and turn-on specifications. **Hours 25-26.** Pull resistors and logic-level compatibility table. **Hours 27-28.** Flyback diode, return currents and decoupling. **Hours 29-30.** Wires, connectors, crimping and protection selection for the actual circuit.

Hours 31-40 - practice. 2 h: pull-up switch and RC. 2 h: inductive-switch schematic, simulation and component checks. 2 h: physical assembly of the checked circuit. 2 h: measurements, static turn-off and log. 2 h: pinout, unpowered connection checks and verification report.

Intermediate acceptance: another person can use the schematic to trace on/off current paths, identify flyback-diode polarity and find protection ratings. Documentation supports all voltages and permissible operating conditions. If hardware is not ready, software block C proceeds independently; physical items remain open without being falsely marked complete.

Work sequence: hours 41-75

Block C: record a signal and verify its interpretation

Hours 41-42. Sampling, fs and Nyquist; predict an alias frequency by hand. **Hours 43-44.** Quantisation, saturation, timestamps and jitter. **Hours 45-46.** Low-pass, high-pass and notch filters, and causality. **Hours 47-48.** Moving average and IIR; calculate state after one sample. **Hours 49-50.** Planar complementary filter and accelerometer-angle assumptions.

Hours 51-60 - practice. 2 h: 70→30 Hz aliasing and filtering before downsampling. 2 h: ADC model and missing sample. 2 h: causal encoder filters. 2 h: complementary angle and body acceleration. 2 h: error table and restart/NaN tests. Use the same reference when comparing settings.

By hour 60, implement streaming interface update(sample,timestamp). It updates state without future data and returns a value plus diagnostic flag. A separate saved-file analysis may use the whole recording, but its result name and delay conclusions must reflect that.

Block D: frequencies, latency and presentation

Hours 61-62. FFT, windows, resolution and PSD. **Hours 63-64.** Frequency response, bandwidth and phase shift. **Hour 65.** Build the ADC/sensor/transport/filter/PWM latency budget and identify unknown times.

Hours 66-75 - practice. 2 h: spectrum and Welch parameters. 2 h: sinusoidal runs and freqz. 2 h: final noise, error and delay table. 2 h: six error checks and reproduction. 2 h: assemble electronics log and report, give an oral explanation, and list M06 hardware questions.

Six required failure cases: incorrect polarity detected before power-up; mirrored pinout found by continuity checks; ADC range excess identified by calculation and simulation; aliasing not 'fixed' by later filtering; NaN does not permanently corrupt IIR; a long gap is flagged and handled by the chosen policy. The first three require no deliberate damage to the physical test bench.

Final new variant: change fs by a factor of two and, before running, predict changes in an 11-sample window's delay, the IIR coefficient for the same time constant and the observed alias frequency. Verify the calculation. If the programme simply keeps the old coefficients, explain why filter behaviour in physical seconds changes.

Acceptance: a measured circuit and an explainable filter

Deliverable package

The electronics log includes schematics and revisions, selected components and documentation, resistance/power/protection calculations, probe positions, assembly photographs and measurement tables. The switching circuit shows current paths, flyback diode, turn-off and measurement limitations. Connections include pinout, labels and confirmed unpowered continuity checks.

The signal-processing report includes a reference generator, real M03 recordings, fs metadata and timestamps, causal filters, settings, time plots and spectra, noise/delay metrics, gap handling and tests. Do not present a smoothed real-IMU signal as unknown ground truth; calculate reference RMSE on synthetic data, and assess repeatability and observable properties on real data.

Questions to explain

Why are voltmeters and ammeters connected differently? How does loading change divider output? Why does an LED need current limiting? How does VGS(th) differ from a low-RDS(on) operating condition? Where does coil current flow immediately after turn-off? Why is the common return part of the signal path? How are fuse parameters selected, and why do they depend on wiring?

How does sampling rate differ from PWM frequency? Why cannot digital filtering restore information lost through aliasing? What changes when the window grows? How does step delay differ from rise time? Why is filtfilt not a streaming filter? How does body acceleration distort a simple accelerometer-angle estimate? What are PSD units?

Completion criteria

Passing work includes a correctly assembled and protected low-power circuit after checking, correct multimeter measurements, recognised aliasing, comparison of at least two causal filters and a quantified delay tradeoff. Label physical experiments and simulations separately. Fill blank templates with your own results; the document does not claim accuracy in advance.

Pass M06 the pinout, signal ranges, expected rates, error handling and low-voltage assembly log. M07 uses current calculations and understanding of inductive loads, but its power bench requires its own components, protection and tests after G1. State one sensor-path limitation each for the leg and hand; for example, permissible delay remains undefined until the future loop dynamics are known.

Assigned reading: a short route

Read only the specified passages before the corresponding experiment. References are mainly in English; keep explanations, glossary and report in English. Reading and preparation fit within the existing 75 hours; whole books are not required.

Core reading

1. Tony R. Kuphaldt, Lessons in Electric Circuits. DC volume: chapters 1-3, 6, 16. Selected topics: voltage/current, Ohm's law, measurement, dividers, Kirchhoff's laws and RC transients. Before tasks 1-4, draw instrument connections and calculate the circuit. For task 5, read Semiconductors volume, chapter 3, Inductor Commutating Circuits.

2. Steven W. Smith, The Scientist and Engineer's Guide to Digital Signal Processing. Chapter 3: sampling; chapter 15: moving average. For tasks 7 and 9, compare aliasing, noise and delay. In chapter 15, read Noise Reduction vs. Step Response and Frequency Response; a full DSP course is unnecessary.

3. Anant Agarwal, MIT 6.002: Circuits and Electronics, notes. Lectures 2, 4, 12: Basic circuit analysis method; The digital abstraction; Capacitors and first-order systems. Consult after a simple experiment to understand logic-level tolerances and finite RC settling time.

Documentation for individual experiments

Switching and protection. TI ULN2003A: pinout, typical circuit, power supply or Nexperia AN11158: MOSFET parameters. Read the chosen option for task 5: limits, losses and flyback diode. Every circuit component contributes to permissible operation.

Filtering. Analog Devices: Anti-Alias Filters; SciPy freqz and welch. For tasks 7 and 11, specify fs, window, nperseg and scaling; assess causality and units first.

Bus pull-up reference. TI SLVA689, I2C pull-up calculation. Understand resistance, capacitance and edge-rate relationships; build a real I2C bus in M06.

Understanding check. Before assembly, explain current paths when the switch turns on and off; before filtering, identify information already lost during sampling and the cost of noise reduction.

Glossary: terms and definitions

Voltage. Electric potential difference between two points; measured in volts and always requires a reference point.

Current. Rate of electric-charge transfer through a cross-section; measured in amperes, with sign determined by the chosen direction.

Resistance. Voltage-to-current ratio of an ohmic element under specified conditions; measured in ohms and limits circuit current.

Power. Rate of energy transfer; for an electrical circuit, $P=U \cdot I$ with consistent direction conventions, measured in watts.

Ground. Chosen circuit reference node and current-return path; its conductors have nonzero resistance and inductance.

Voltage divider. Series circuit producing a fraction of input voltage; connected loads change the division ratio and must be included.

Time constant. For a simple RC circuit, $\tau=R \cdot C$ in seconds; during τ, step response covers approximately 63% of its full change.

Inductance. Parameter relating current change to voltage; a coil opposes rapid current changes and stores magnetic energy.

Flyback diode. Diode providing a coil-current path after the switch opens; supply configuration determines its polarity and ratings.

MOSFET. An insulated-gate field-effect transistor; gate-source voltage controls channel conductance, and switching requires changing gate charge.

Pull-up. Resistor between signal line and supply that establishes a high level when the input or open drain is high impedance. At a low level, it limits current.

Fuse. Component that breaks a circuit under a specified current-time load; protects the selected wiring path within its rated characteristics.

Sampling. Representing a continuous signal by samples in time; $f_s=1/\Delta t$ on a uniform time grid.

Aliasing. Frequency ambiguity after sampling in which different continuous signals produce identical samples, causing information loss.

Quantisation. Replacing a continuous value with one of a finite set of levels; introduces amplitude error distinct from temporal sampling.

Causal filter. Filter using only present and past inputs; future samples are unavailable during streaming.

FIR and IIR. FIR has a finite impulse response; IIR usually uses feedback and may respond for an unlimited duration.

Power spectral density (PSD). Distribution of signal power over frequency; integrating a band gives its mean-square contribution under consistent normalisation.

Complementary filter. Combination of estimates with complementary frequency responses, such as slow accelerometer tilt and rapid gyroscope angle changes.

Equipment, installation and checks

Complete preparation within the first study blocks in place of another general overview. The budget remains 75 hours; delivery waiting time is excluded. Statuses below specify actions for you to take, not purchases or installations already completed.

Use existing equipment

Use the MacBook Pro M4 and Python/SciPy from M02-M03, plus sensor recordings or synthetic data for filtering. Reuse listed tools if already available and working.

Buy before the first circuit experiment

1 each: a multimeter measuring DC voltage and resistance, with continuity, diode test and a fused current input; a set of working probes. Supplement this measurement guide with your model's manual. Connect across a circuit for voltage and in series for current.

1 CC/CV bench supply with adjustable voltage, current limit and independent output switching. A guideline for the shared M04-M07 bench is approximately 0-30 V and up to 5-10 A at the required voltage. These are supply capabilities; calculate each experiment's low-voltage, low-power operating point. Do not route full supply power through a breadboard.

1 low-power circuit kit: breadboard and jumpers; several each of resistors from 330 ohm to 100 kohm, LEDs and buttons; 100 nF and 10-100 µF capacitors; 1 low-power, low-voltage relay with unloaded contacts, 1 compatible switch and flyback diode. Select final values, powers and voltages from tasks 1-5 and documentation before ordering.

1 connection kit: adjustable soldering iron or station with stand, solder, flux and fume extraction; side cutters, wire stripper, tool for the chosen crimp contacts, wires, connectors, heat-shrink, holder and suitable fuses. Consumable quantities should cover one bench and task 6 samples.

Install and check now

macOS ARM: LTspice for macOS for RC and switching circuits; verify your macOS version and Apple Silicon support in the selected release. SciPy.signal is already available. Test: RC step response reaches about 63% at t=RC, freqz displays a Hz axis, and welch reproduces after restart.

Before power-up: continuity and polarity checks, correct multimeter sockets, calculated supply limits and protection ratings; then test a low-power resistive load according to the supply manual. M04 needs neither a motor nor MCU firmware. An oscilloscope remains optional: simulate fast transients and explicitly state that no physical measurement was made.

Motion fundamentals: what a sampled sensor sees

Compulsory, within 75 hours. Use 1 h from 63-64 and 2 h of practice 66-75, 'sinusoidal runs and freqz'. Use existing generators and the filter from tasks 7, 9 and 11; this is their combined physical interpretation. Before starting, calculate $\Delta t=1/100=0.01$ s and convert 20 ms to seconds.

Motion continues between samples

Let a joint in a fixed plane have angle $q(t)=0.20\sin(2\pi\cdot 2t)$ rad relative to the horizontal axis. Its velocity $\dot{q}=0.80\pi\cos(4\pi t)$ rad/s reaches 2.513 rad/s. For link $L=0.12$ m, maximum endpoint speed $L|\dot{q}|\approx 0.302$ m/s. The sensor measures q, not foot velocity directly; calculation supplies geometry and differentiation.

The example recording also contains sinusoidal interference $e(t)=0.01\sin(2\pi\cdot 70t)$ rad. At fs=100 Hz, its samples match $-0.01\sin(2\pi\cdot 30k/100)$: 70 Hz appears as 30 Hz with a different phase. This sequence cannot establish the original continuous oscillation. A later filter may attenuate 30 Hz but cannot restore lost information; antialiasing is needed before sampling or reducing its rate.

Why a smoother angle may arrive too late

Consider a causal five-sample average: $q_f[k]=(q_m[k]+\ldots+q_m[k-4])/5$, where q_m is measured angle. Output is issued at $k\Delta t$ although the window centre is two samples earlier. In steady state at 2 Hz, delay is $(5-1)\Delta t/2=0.020$ s and phase shift is $-2\pi\cdot 2\cdot 0.020=-0.2513$ rad, or $-14.4°$.

The amplitude of the desired sinusoid is multiplied by $\sin(5\pi f/f_s)/(5\sin(\pi f/f_s))\approx 0.98428$ at f=2 Hz. Thus the smoothed angle nearly preserves amplitude but describes earlier motion. Treating it as current gives endpoint position error of scale $|v|\cdot 0.020\approx 6$ mm during rapid motion. This is a local delay-effect estimate, not an accuracy tolerance for a finished leg.

Checks for the future control loop

1. Compare sampled 70 Hz interference with a negative 30 Hz sinusoid; repeat at fs=250 Hz. Save overlaid sequences and label true frequency only where known from the generator. A measured spectral peak does not establish a mechanical cause.

2. On clean 2 Hz motion, measure the five-sample filter's amplitude and phase after the transient and compare with freqz. For a unit step with zero prehistory, output first exceeds 0.5 at 20 ms after the first changed input sample. Add these values to the task 11 table; account for transport delay separately.

Do not shift streaming data backwards in time to claim less delay: an earlier timestamp does not speed publication. Feedback suitability depends on the future M10 loop dynamics, not plot smoothness. All motion here is synthetic; no motor is needed.

Read before calculating: Steven W. Smith, DSP Guide: chapter 3, The Sampling Theorem; chapter 15, Noise Reduction vs. Step Response.